INCH-POUND

MIL-I-275D
27 February 1990
SUPERSEDING
MIL-F-275C
13 December 1982

MILITARY SPECIFICATION

IRON OXIDE, BLACK, TECHNICAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope**. This specification covers four classes of technical grade black iron (II, III) oxide (Fe_3O_4).

1.2 **Classification**. Iron oxide shall be of the following classes as specified (see 6.2):

Class A - 80 mesh
Class B - 12 mesh
Class C - 100 mesh
Class D - 200 mesh

1.3 **Part or identifying number (PIN)**. A specification-based PIN to identify the classes of iron oxide is included in section 6.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards, and handbooks**. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-PET-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6810

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otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

RR-S-366 - Sieve Test

STANDARDS

MILITARY

MIL-STD-1168 - Ammunition Lot Numbering

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issue of the documents cited in the solicitation (see 6.2).

ASTM STANDARDS

D 1193 - Reagent Water
D 3872 - Ferrous Iron in Iron Oxides
D 3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 **Chemical characteristics.** Iron oxide shall conform to the chemical characteristics of table I when tested as specified therein.

TABLE I. **Chemical characteristics**

Characteristic	Percent by weight		Test paragraph
	Minimum	Maximum	
Moisture	----	0.20	4.2.4.1
Metallic iron	----	1.00	4.2.4.2
Ferrous ion (Fe++)	22.0	35.9	4.2.4.3
Total iron	69.0	----	4.2.4.4

3.2 **Granulation characteristics.** Iron oxide shall conform to the applicable granulation characteristics of table II when tested as specified in 4.2.4.5.

TABLE II. **Granulation characteristics**

Characteristic	Percent by weight passing			
	Class A	Class B	Class C	Class D
No. 12, minimum	----	99.9	----	----
No. 20, minimum	----	95	----	----
No. 50, maximum	----	70	----	----
No. 80, minimum	99.0	----	----	----
No. 100, minimum	----	----	99.0	----
No. 140, maximum	----	10	----	----
No. 200, minimum	----	----	----	99.0
No. 270, maximum	----	----	50	----

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set

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forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the iron oxide of one class, produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 **Sampling.**

4.2.2.1 For examination of packaging. Sampling shall be conducted as specified in the contract (see 6.2).

4.2.2.2 For test. Sampling shall be conducted in accordance with table III. A representative specimen of approximately 1 pound shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

TABLE III. **Sampling for test**

Number of containers in batch or lot	Number of sample containers
3 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

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4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects:

(a) Contents per container not as specified
(b) Container damaged or leaking
(c) Unitization not as specified
(d) Marking incorrect, missing, or illegible

4.2.3.2 For test. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.4 Tests. Water in accordance with ASTM D 1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

4.2.4.1 Moisture. Heat a moisture dish in an oven at 105° to 110°C for 1 hour, cool in a desiccator, and weigh to the nearest 0.1 milligram (mg). Transfer approximately 10 grams (g) of the specimen to the dish, stopper, and weigh to the nearest 0.1 milligram. Heat unstoppered in an oven at 105° to 110°C for 1 hour, cool to room temperature in a desiccator, replace the stopper, and weigh to the nearest 0.1 mg. Repeat the heating for 1/2-hour periods until successive weighings differ by no more than 0.1 mg, but do not continue heating for more than 3 hours. Calculate the loss in weight as percent by weight moisture as follows:

$$\text{Percent by weight moisture} = \frac{100(A - B)}{(A - C)}$$

where: A = Weight of dish and specimen before heating, in grams,
B = Weight of dish and specimen after drying to constant weight, in grams, and
C = Weight of empty, dry dish, in grams.

4.2.4.2 Metallic iron.

(a) **Zimmerman-Reinhardt reagent.** Dissolve 51 g of manganous sulfate, monohydrate ($MnSO_4 \cdot H_20$) in 300 milliliters (mL) of water. Add a cooled solution of 130 mL of concentrated sulfuric acid in 300 mL of water and 138 mL of concentrated phosphoric acid. Dilute to 1 liter with water.

(b) **Procedure.** Grind 3 to 4 g of the specimen to pass a No. 40 sieve. Weigh to the nearest 0.1 mg approximately 2 g of the ground specimen into a 250-mL Erlenmeyer

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flask. Add 3 g of mercuric chloride, displace the air from the flask with a continuous flow of carbon dioxide and add 100 mL of boiling water. Place on a steam bath for 10 minutes and agitate frequently while continuing the flow of carbon dioxide. Remove from the steam bath, cool to room temperature, and discontinue the flow of carbon dioxide. Filter, and wash the residue with water, catching the filtrate and washings in a beaker containing 25 mL of Zimmerman-Reinhardt reagent prepared as specified in (a) and 350 mL of water. Titrate immediately with 0.1*N* potassium permanganate solution. Calculate the percent by weight metallic iron as follows:

$$\text{Percent by weight metallic iron} = \frac{5.585AB}{W}$$

where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen, in grams.

4.2.4.3 Ferrous ion. Determine the ferrous ion content of the specimen in accordance with ASTM D 3872.

4.2.4.4 Total iron. Grind approximately 1 g of the specimen to pass a No. 40 sieve. Weigh to the nearest 0.1 mg approximately 0.2 g of the ground specimen into a porcelain crucible weighed to the nearest 0.1 mg. Ignite at dull red heat for at least 1 hour. Cool to room temperature, add 5 mL of concentrated hydrochloric acid, cover with a watch glass, and place on a steam bath. As the characteristic yellow color of ferric chloride forms, decolorize by adding dropwise, and with stirring, a solution made by dissolving 50 g of stannous chloride, dihydrate ($SnCl_2 \cdot 2H_20$) in 100 mL of concentrated hydrochloric acid and diluting to 1 liter with water. (Note: A large excess of stannous chloride leads to the formation of black compounds which spoil the analysis, or to excessive amounts of mercurous chloride which leads to high results. If the proper amount of stannous chloride has been added, a slightly silky precipitate of mercurous chloride is obtained.) Continue the dropwise addition of the stannous chloride solution as the ferric chloride color forms until all of the specimen is dissolved and the solution is completely decolorized. Add one or two drops of the stannous chloride solution in excess. Remove from the steam bath, cool to room temperature, and add at one stroke, 15 mL of 10-percent mercuric chloride solution. Allow to stand for 2 minutes covered with a watch glass. Wash the contents of the crucible into a beaker containing 25 mL of Zimmerman-Reinhardt reagent prepared as specified in 4.2.4.2(a) and 400 mL of water. Titrate immediately with 0.1*N* potassium permanganate solution. Calculate the percent by weight total iron as follows:

$$\text{Percent by weight total iron} = \frac{5.585AB}{W}$$

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where: A = Milliliters of potassium permanganate solution used,
B = Normality of potassium permanganate solution, and
W = Weight of specimen, in grams.

4.2.4.5 Granulation characteristics. Select sieves conforming to RR-S-366 as specified in table II. Weigh the dry sieves and a bottom pan. If more than one sieve is specified, nest the sieves in order of increasing fineness with the coarsest sieve on top. Place the nest of sieves on the bottom pan. Weigh to the nearest 0.01 g approximately 100 g of the specimen and transfer to the top sieve. Cover and shake for 3 minutes by means of a mechanical shaker geared to produce 300 ± 15 gyrations and 150 ± 10 taps of the striker per minute. Weigh the specimen retained on each sieve and on the bottom pan and calculate the percent specimen passing through each sieve.

5. PACKAGING

5.1 Unit packing, industrial. A quantity of 50 (+1/2 or -0) pounds of iron oxide shall be unit packed in accordance with ASTM D 3951.

5.2 Packing. Iron oxide, unit packed as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Unitization. Iron oxide, packed as specified above, shall be unitized in a manner to assure acceptance by the common carrier applicable to the mode of transportation, safe delivery to destination, and stable stacking in storage.

5.4 Marking. Containers shall be marked in accordance with ASTM D 3951. Unitized loads shall be marked to assure arrival at destination.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Iron oxide is intended for the uses listed in table IV.

TABLE IV. **Intended uses**

Classification	For use in the manufactire of
Class A	Pyrotechnic mixtures such as CN candle and starter mixures
Class B	Thermite
Class C	First fire mixtures
Class D	Starter mixtures

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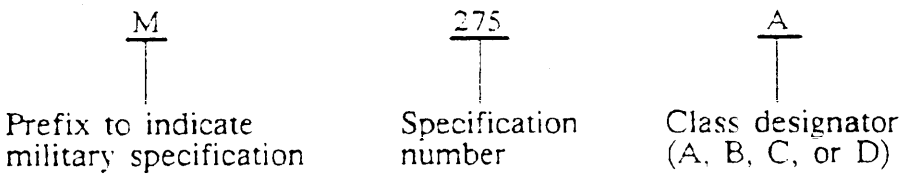
6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)
(c) Class of iron oxide required (see 1.2)
(d) Sampling for examination of packaging (see 4.2.2.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E 29.

6.5 Part or identifying number (PIN). The PINs to be used for items acquired to this specification are created as follows:



6.6 Subject term (key word) listing.

Ferrous oxide
Metallic iron
Total iron

Custodians:

Army - EA
Navy - OS

Review activities:

Army - AR, MD, MI
Navy - AS
DLA - GS

Preparing activity:

Army - EA

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-275D	IRON OXIDE, BLACK, TECHNICAL

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION (Mark one)
	☐ VENDOR
	☐ USER
b. ADDRESS (Street, City, State, ZIP Code)	☐ MANUFACTURER
	☐ OTHER (Specify): _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER (Last, First, MI) – Optional	b. WORK TELEPHONE NUMBER (Include Area Code) – Optional
c. MAILING ADDRESS (Street, City, State, ZIP Code) – Optional	8. DATE OF SUBMISSION (YYMMDD)

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